

Question 1

(a) We have that

$$\begin{aligned}\{f_n > t\} &= \left\{x \in \mathbb{R} : f\left(\frac{[nx]}{n}\right) > t\right\} = \left\{x \in \mathbb{R} : \frac{[nx]}{n} \in \{f > t\}\right\} = \{x \in \mathbb{R} : [nx] \in n\{f > t\}\} \\ &= \{x \in \mathbb{R} : \exists k \in n\{f > t\}, [nx] = k\}\end{aligned}$$

But for any $k \in n\{f > t\}$, we have that $[nx] = k \iff k \in \mathbb{Z}$ and $k \leq nx < k+1 \iff k \in \mathbb{Z}$ and $x \in [\frac{k}{n}, \frac{k+1}{n})$. So

$$\{f_n > t\} = \bigcup_{\substack{k \in n\{f > t\} \\ k \in \mathbb{Z}}} \left[\frac{k}{n}, \frac{k+1}{n}\right)$$

(b) Each $[\frac{k}{n}, \frac{k+1}{n}) \in \mathcal{B}(\mathbb{R})$, so $\bigcup_{\substack{k \in n\{f > t\} \\ k \in \mathbb{Z}}} [\frac{k}{n}, \frac{k+1}{n}) \in \mathcal{B}(\mathbb{R})$, as a countable union so f_n is Lebesgue measurable

(c) We have that $\frac{[nx]}{n} \leq \frac{nx}{n} = x$. So the sequence $\left(\frac{[nx]}{n}\right)_{n \in \mathbb{N}}$ is an increasing sequence converging to x . So, since f is continuous from the left, we have that

$$\lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} f\left(\frac{[nx]}{n}\right) = \lim_{y \rightarrow x^-} f(y) = f(x)$$

Then, since each f_n is Lebesgue measurable, f is as a limit of Lebesgue measurable functions.

Question 2

(a) We have that

$$\int_{(1,\infty)} |f|^p d\lambda_1 = \int_{\mathbb{R}} |f|^p \chi_{(1,\infty)} d\lambda_1 = \int_{\mathbb{R}} \lim_{n \rightarrow \infty} (\chi_{(1,n)} |f|^p) d\lambda_1$$

The function $|f|^p \chi_{(1,n)}$ is increasing, positive, and measurable ($|f|^p L$ is measurable since it is continuous on $(1,n)$, $\chi_{(1,n)}$ is measurable since $(1,n)$ is a Borel set, so $|f|^p \chi_{(1,n)}$ is measurable as a product of measurable function) so by the Monotone Convergence Theorem:

$$\int_{(1,\infty)} |f|^p d\lambda_1 = \lim_{n \rightarrow \infty} \int_{\mathbb{R}} |f|^p \chi_{(1,n)} d\lambda_1 = \lim_{n \rightarrow \infty} \underbrace{\int_{(1,n)} |f|^p d\lambda_1}_{f \text{ is bounded on } (1,n) \text{ so Lebesgue} = \text{Riemann}} = \lim_{n \rightarrow \infty} \int_1^n |f|^p dx = \int_1^\infty |f|^p dx$$

So we can consider the improper Riemann integral and use the Comparison Test. Since $1 + |\ln(x)| \geq 1$, we get

$$\frac{1}{(1 + |\ln(x)|)^p x^{p/2}} \leq \frac{1}{x^{p/2}}$$

Now

$$\int_1^\infty \frac{1}{x^{p/2}} dx \text{ converges} \iff \frac{p}{2} > 1 \iff p > 2$$

So, for $p > 2$, we get that $f \in L^p(1, \infty)$. On the other hand, we can write $(1, \infty) = \bigcup_{n \in \mathbb{N}^*} (e^n, e^{n+1}]$. So, if we can show

$$\lim_{n \rightarrow \infty} \int_1^{e^{n+1}} \frac{1}{(1 + |\ln(x)|)^p x^{p/2}} dx = \lim_{n \rightarrow \infty} \sum_{k=0}^n \int_{e^k}^{e^{k+1}} \frac{1}{(1 + |\ln(x)|)^p x^{p/2}} dx$$

diverges, then so will $\int_1^\infty \frac{1}{(1 + |\ln(x)|)^p x^{p/2}} dx$. On each interval $(e^k, e^{k+1}]$, $\ln x \leq k+1$ and $x \leq e^{k+1}$. So,

$$\sum_{k=0}^n \int_{e^k}^{e^{k+1}} \frac{1}{(1 + |\ln(x)|)^p x^{p/2}} dx \geq \sum_{k=0}^n \int_{e^k}^{e^{k+1}} \frac{1}{(k+2)^p e^{\frac{p(k+1)}{2}}} dx = \sum_{k=0}^n \frac{e^k(e-1)}{(k+2)^p e^{\frac{p(k+1)}{2}}} \geq \sum_{k=0}^n \frac{1}{(k+2)^p e^{\frac{1}{2}(pk+p-2k)}}$$

If $p < 2$, then $pk + p - 2k = k(p-2) + p$, which, for k large enough, is negative. Then,

$$\lim_{k \rightarrow \infty} \frac{1}{(k+2)^p e^{\frac{1}{2}(pk+p-2k)}} = \infty$$

so this series diverges, meaning $f \notin L^p(1, \infty)$

(b) We have that

$$\int_1^\infty \frac{1}{(1 + \ln x)^2 x} dx = \lim_{n \rightarrow \infty} \int_1^n \frac{1}{(1 + \ln x)^2 x} dx$$

Letting $u = \ln x$, we get

$$\int_1^n \frac{1}{(1 + \ln x)^2 x} dx = \int_0^{\ln n} \frac{1}{(1 + u)^2} dx = -\frac{1}{1 + u} \Big|_0^{\ln n} = 1 - \frac{1}{1 + \ln n} \rightarrow 1 \text{ as } n \rightarrow \infty$$

So $f \in L^2(1, \infty)$

(c) Again, with the same reasoning as in (a), let (a_n) be any decreasing sequence converging to 0. Then we have

$$\int_{(0,1)} |f|^p d\lambda_1 = \int_{\mathbb{R}} |f|^p \chi_{(0,1)} d\lambda_1 = \int_{\mathbb{R}} \lim_{n \rightarrow \infty} (|f|^p \chi_{(a_n,1)}) d\lambda_1 = \lim_{n \rightarrow \infty} \int_{(a_n,1)} |f|^p d\lambda_1 = \lim_{n \rightarrow \infty} \int_{a_n}^1 |f|^p dx$$

Since this is true for any sequence (a_n) , we get

$$\int_{(0,1)} |f|^p d\lambda_1 = \int_0^1 |f|^p dx$$

Again, we can consider the improper Riemann integral and use comparison tests.

Again, $\frac{1}{(1 + |\ln(x)|)^p x^{p/2}} \leq \frac{1}{x^{p/2}}$. Now

$$\int_0^1 \frac{1}{x^{p/2}} dx = \lim_{a \rightarrow 0^+} \int_a^1 x^{-p/2} dx = \begin{cases} \left[\ln(x) \right]_a^1 & \text{if } p/2 = 1 \\ \left[\frac{1}{1-p/2} x^{1-p/2} \right]_a^1 & \text{if } p/2 \neq 1 \end{cases}$$

Then $\ln x \rightarrow -\infty$ as $a \rightarrow -\infty$. On the other hand, if $1 - p/2 > 0$, then $x^{1-p/2}$ converges as $x \rightarrow 0^+$ (and diverges otherwise). That is, $f \in L^p(0, 1)$ if $p < 2$. For $p = 2$, we get

$$\int_0^1 \frac{1}{(1 + \ln x)^2 x} dx = \lim_{a \rightarrow 0^+} \int_a^1 \frac{1}{(1 + \ln x)^2 x} dx$$

Letting $u = \ln x$, we get

$$= \int_{\ln a}^1 \frac{1}{(1 + u)^2} dx = -\frac{1}{1 + u} \Big|_a^1 = \frac{1}{1 + a} - \frac{1}{2} \rightarrow \frac{1}{2} \text{ as } a \rightarrow 0^+$$

So $f \in L^2(0, 1)$

On $(0, 1)$, for $p > 2$ we have that

$$\frac{1}{(1 + |\ln x|)^{p/2}} \geq \frac{1}{(1 + |\ln x|)^p} \geq \frac{1}{1 + |\ln x|} = \frac{1}{1 - \ln x}.$$

Noticing that $(0, 1) = \bigcup_{n \in \mathbb{N}^*} (1 - e^{-n}, 1 - e^{-n-1}]$ and doing the same as in (b), we get

$$\lim_{n \rightarrow \infty} \sum_{k=0}^n \int_{1-e^{-k}}^{1-e^{-k-1}} \frac{1}{1 - \ln x} dx \geq \lim_{n \rightarrow \infty} \sum_{k=0}^n \int_{1-e^{-k}}^{1-e^{-k-1}} e^k dx = \lim_{n \rightarrow \infty} \sum_{k=0}^n e^{-k-1} (e - 1) e^k$$

And the terms in the sum $e^{-k-1} e^k (e - 1) = 1 - e^{-1} \not\rightarrow 0$ so this sum diverges. Thus $f \notin L^p(0, 1)$ for $p > 2$

- (d) $f \in L^p(0, \infty) \iff \int_0^\infty |f|^p dx < \infty$. Since we can write $\int_0^\infty |f|^p dx = \int_0^1 |f|^p dx + \int_1^\infty |f|^p dx$, it follows $f \in L^p(0, \infty) \iff f \in L^p(0, 1) \cap L^p(1, \infty)$. Using our results from (a), (b) and (c), $f \in L^p(0, \infty) \iff p = 2$